

Design of a Small Bio-Inspired Jumping Robot for Low-Gravity Exploration

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Abstract

Exploration of alien celestial bodies is a challenge for robots. Obstacles such as large rocks, steep cliffs, vast craters and unpredictable caves are extremely difficult to traverse. This is usually accompanied by limited energy sources (solar) and no atmosphere, which limits aerodynamic capabilities of robots. By jumping, robots can overcome most, if not all these challenges with ease.

Therefore, the aim of this project is to create a small robot that can move by jumping. The robot will have no aerodynamic support (i.e., propellers, wings, etc.), and will be fully electric, meaning no expendable fuel that would make it useless after a few jumps.

Background

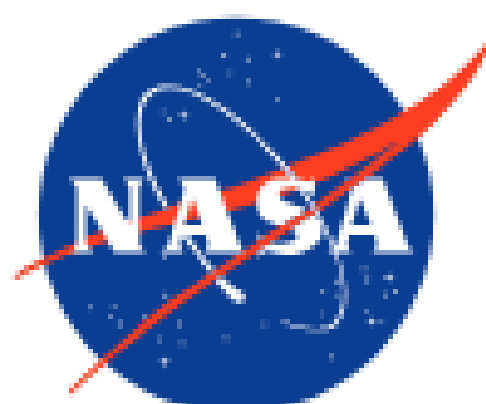
Motivation

Animals adapted for jumping excel at traversing difficult terrain and covering vast distances extremely quickly and efficiently. A perfect example is a Lesser Galago, or 'Bush Baby'. It is incredibly mobile in its natural woodland habitat which is full of uneven terrain and obstacles bigger than itself. A robot that could negotiate terrain like a Bush Baby would be far more capable at exploring the challenging and unpredictable terrain.



Ongoing Investment into the Technology

Jumping robots have been researched for space, military, and search-and-rescue operations. They have been discussed by NASA since 1969 as part of the Apollo mission [1], and continue to see investment from them, as NASA plans to send a bio-inspired jumping robot onto Saturn's moon, Enceladus, in the 2030's as part of the LEAP program [2].



Advantages Over Traditional Rovers

Rovers are limited in what obstacles they can traverse. Their wheel radius determines the largest crevice/rocks they can overcome. They are slow so as to conserve power and avoid tipping. Rovers cannot drive down steep hills and must circumnavigate cliffs. None of these problems apply to jumping robots. Additionally, these robots get more agile the lower the gravity they are in.



Aims

The main aims and goals of this project are as follows:

- Create a small (< 300g) robot that can jump 30-50cm vertically
- The robot must be fully electric with no expendable fuel
- And it must not have aerodynamic supports such as propellers or wings

Robot Design

- The design starts at the base where 2 'ankles' (ball bearings and shafts) connect the 'foot' of the design to a 'tibia'. These are connected via torsion spring at the 'knee' to a 'femur', which itself leads to a 'hip' platform upon which the gearbox lies.
- The robot weighs a total of 296 grammes and is 17cm tall. 119g is the gearbox for the jumping mechanism and 177g are for the legs of the robot.
- A Parallax 900-00008 rated for 0.686Nm of torque drives a gearbox with a reduction of 3.5x. Ideal torque of this system is ~2.4Nm.
- The gearbox drives a 4mm shaft that spools a fishing line. This line is connected to the base of the robot. As it spools, the robot squats down. Elastic tubes between its knees are pulled apart, storing mechanical energy.
- When the motor reverses direction, a one-way bearing forces the motor-driven gears to decouple from the spool, allowing it to freewheel, rapidly releasing all the stored energy. The energy is released into the ground, causing the robot to jump

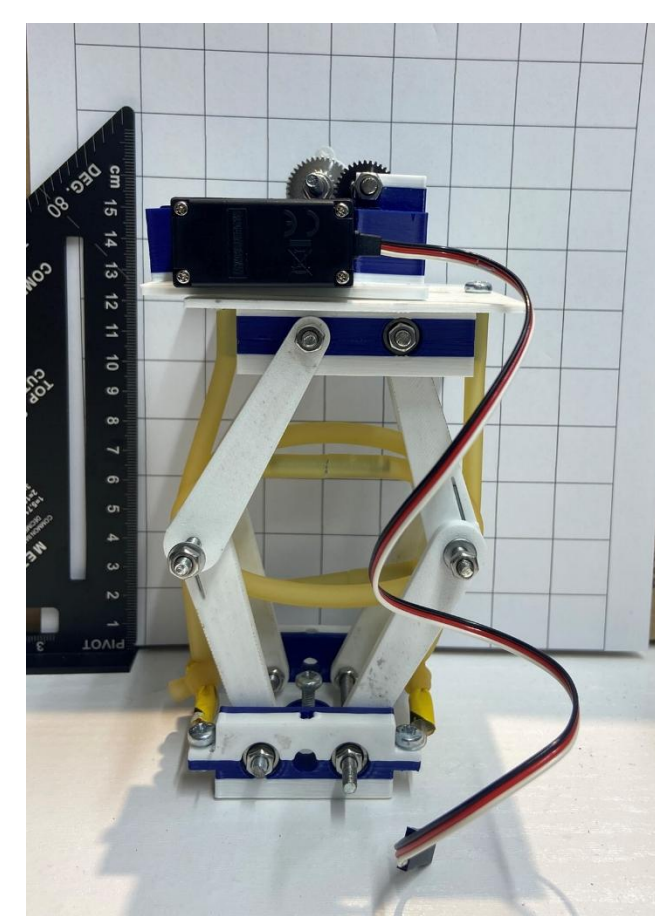


Figure 1. Freestanding robot at rest

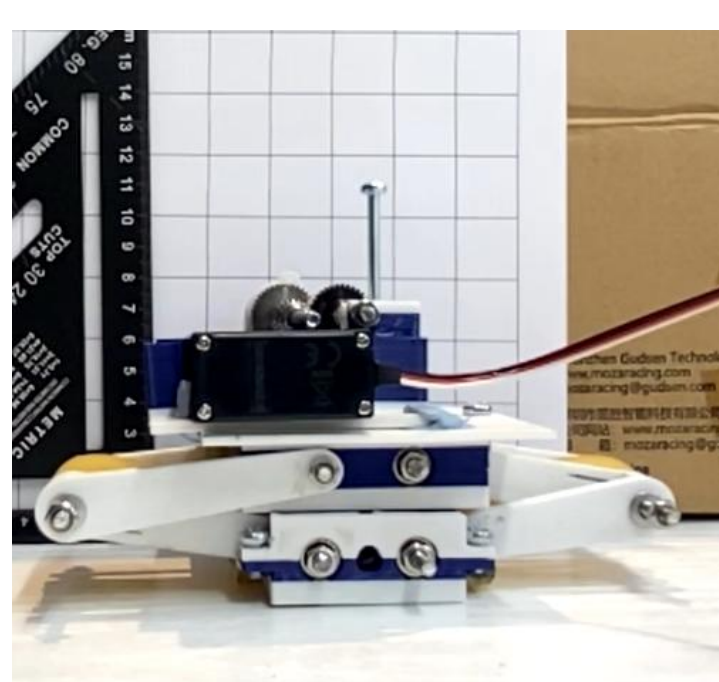


Figure 2. Robot tensioned, ready to jump

Performance

- The robot goes from rest to fully tensioned in approximately 63 seconds.
- Rubber tubing on the robot stores – depending on the length of the tube – between 30N and 36N of force when fully tensioned per tubing run.
- When using three runs of rubber tubing, the robot achieved a max vertical jump of 26cm.

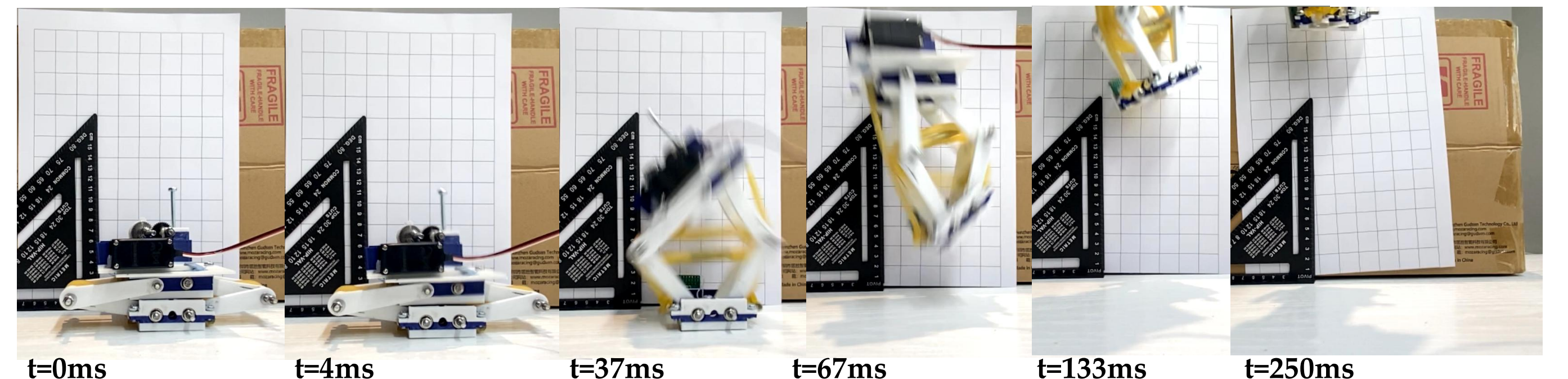


Figure 3. Robot jump showcase from fully tensioned to reaching peak of 26cm (each square on grid is 2x2cm)

- The robot's gearbox is slightly off-centre, shifting the centre of mass and causing the robot to pull to the left when jumping, losing some force that could have otherwise been directed vertically.



Figure 4. QR Code for the above video

Results

- Successfully created a lightweight jumping robot that nearly reached the desired jump height of 30-50cm.

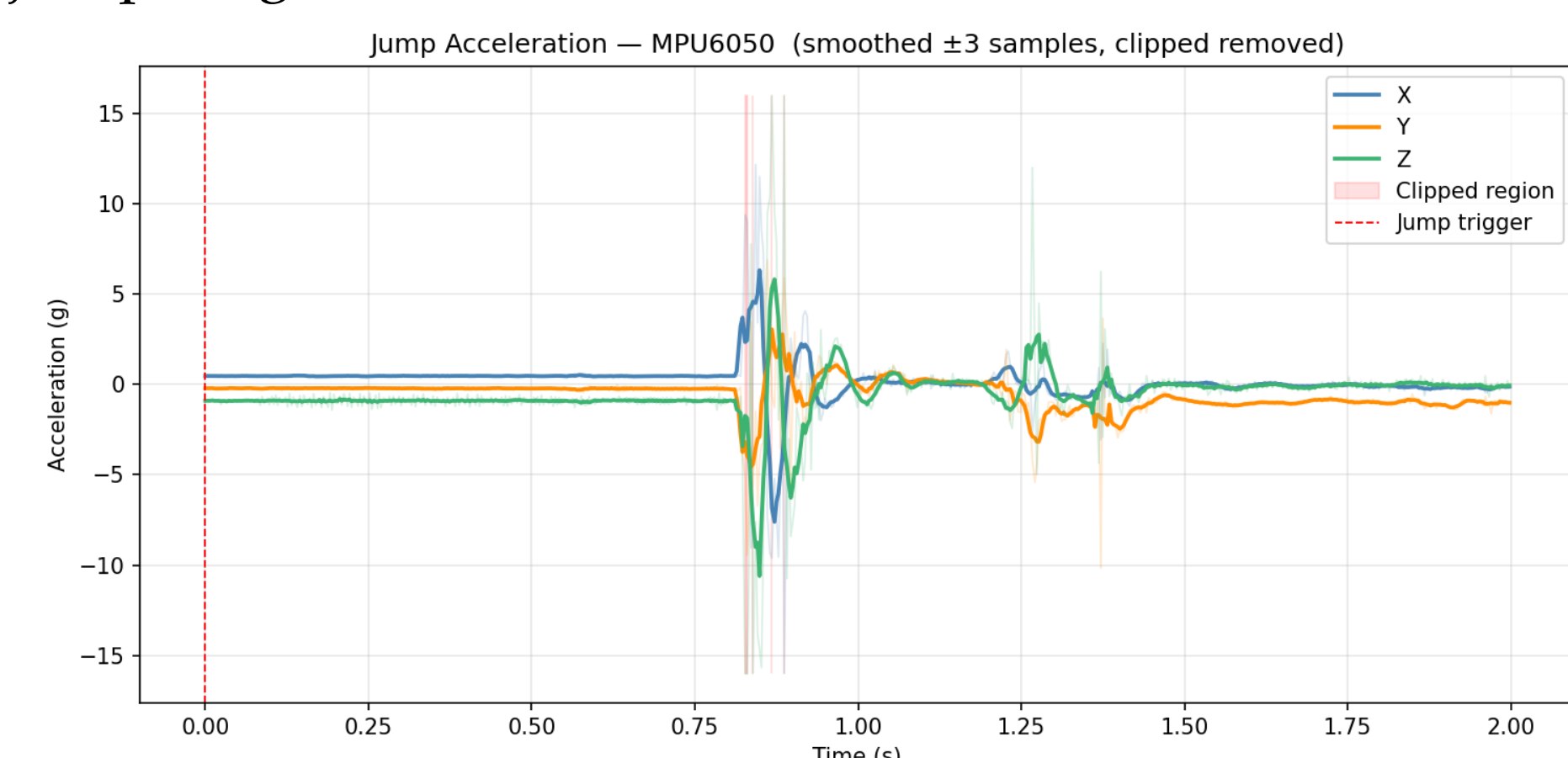


Figure 5. IMU Data for a jump from tensioned to landing

- IMU experienced severe clipping due to instantaneous readings being very high. Moving average of Z-axis acceleration (green line, figure 5) is approximately equal to average acceleration of 6.3g calculated from analysing the slow-motion video (bottom plot, figure 6).

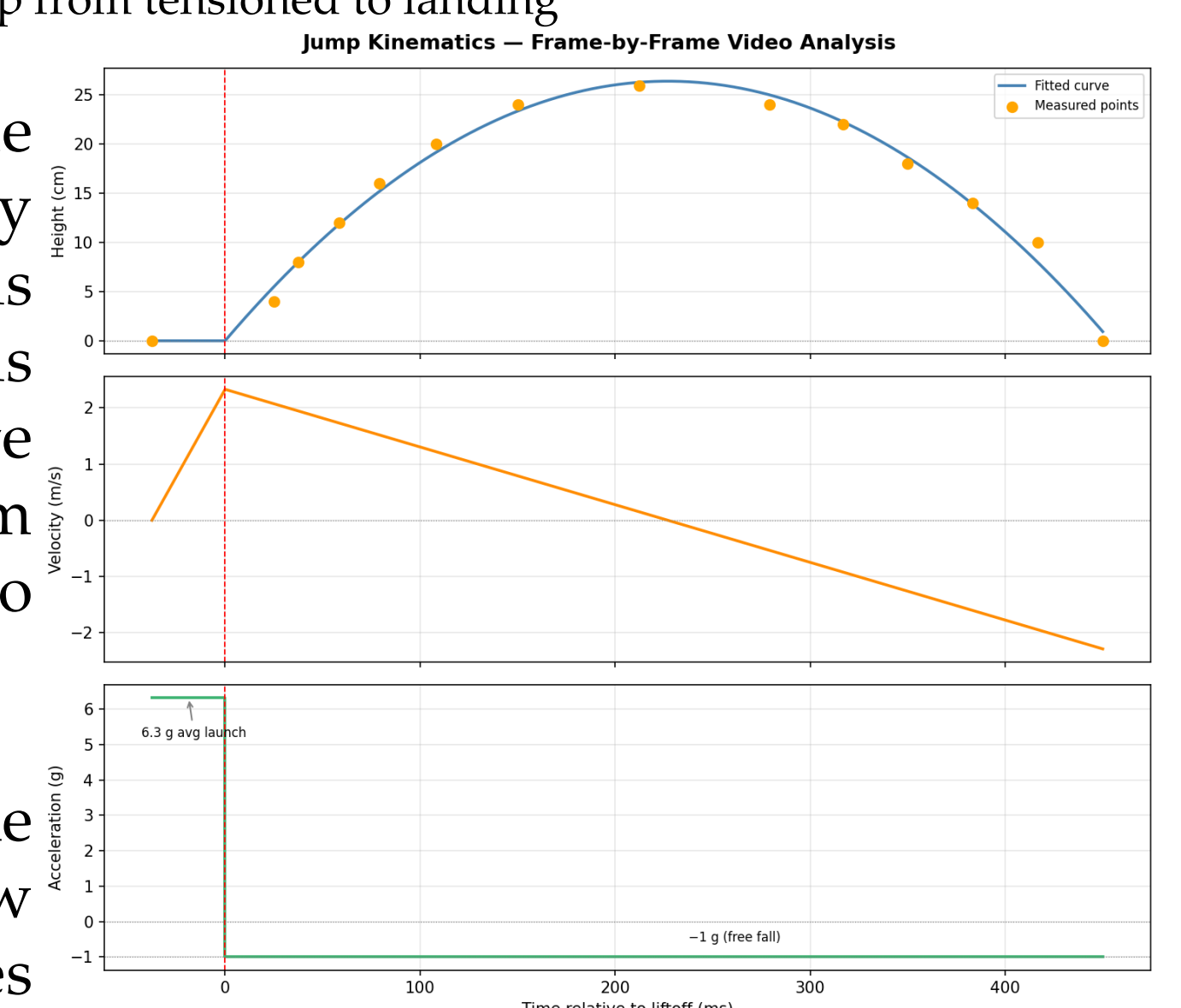


Figure 6. Frame analysis and calculations

- While the robot performed well, the balance is an issue that does not allow for it to jump multiple times consecutively and without intervention

Conclusion

This bio-inspired robot design can successfully jump. Balance is critically important however, without it, the robot cannot land safely or become more agile.

- Improving the efficiency of this design would have it break the 30cm jump height goal with ease. (optimising tubing length, increasing gearbox efficiency, COM improvements)
- This design would greatly benefit from reaction wheels to keep the robot level during flight as well as shifting the center of mass in the design.

References

- [1] Kaplan M.H., Seifert H. "Hopping transporters for lunar exploration. | Journal of Spacecraft and Rockets," *Journal of Spacecraft and Rockets*, 2024, doi: <https://doi.org/10.2514/jsr.1969.6.issue-8>.
- [2] E. Ackerman, "NASA Funds Jumping Robot for Enceladus Exploration," *IEEE Spectrum*, Mar. 03, 2025. <https://spectrum.ieee.org/a-jumping-robot-for-enceladus>

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